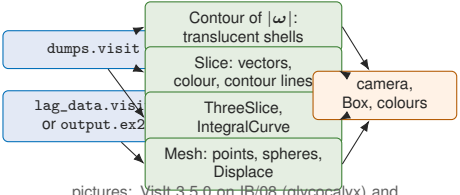
VISIT IN 3D: ISOSURFACES, SLICES, VECTORS AND STREAMLINES

The lectures' "VisIt 3D" recipes redone in VisIt 3.5: the body as points or as a moving mesh, translucent isosurfaces of the vorticity, the camera, vectors and vorticity on a slice, three slices at once, streamlines, and the places where version 3 no longer does what the recipes say



pictures: VisIt 3.5.0 on IB/08 (glycocalyx) and IBFE/10 (plate in a flow tank); the soft coral and the clap-and-fling windows: the 2015 and 2021 decks

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Claude (Anthropic) was used to organize, combine, and format more than fourteen years (2012–2026) of the instructor's tutorial notes from courses and workshops at several universities.

LEARNING GOALS

By the end of this chapter you should be able to:

- Open a 3D run of IBAMR (points or finite elements) and get past the first picture, which is the outside of a box.
- Draw the body in 3D: vertices as pixels or as spheres, or a finite element mesh moved by its displacement.
- Make translucent isosurfaces of the vorticity magnitude with levels, colours and opacity you choose, and frame them with the camera.
- Put vectors, a colour map and contour lines on a slice through the body, and draw three slices at once.
- Draw streamlines with VisIt 3's IntegralCurve operator.
- Recognise the places where VisIt 3.5 no longer does what the lectures' recipes say, and make the whole set of pictures with one script.

> A 3D field shows only its outside until you reduce it: to surfaces (Contour), to planes (Slice, ThreeSlice), to curves (IntegralCurve), or to a box around the body (Box). Every recipe below is one of these reductions plus a camera.

WHERE THIS CHAPTER COMES FROM

- "VISIT 3D" (2015, with Lecture 11; 12 slides): the boundary (pixel size, spheres), isosurfaces of `omega_magnitude`, vectors on a slice, vorticity on the same slice; sample data: a 3D glycocalyx and a 3D leaf (Google Drive folders).
- "VISIT 3D" (2020; 11 slides): the same recipe, without the data, the spheres, "supersaturate" and "lighter to darker".
- "VISIT – How to visualize IBAMR Data" (2021), slides 32–45: the same 3D part with clap-and-fling sample data ("two time steps for two wings performing a clap and fling. Each wing consists of a wing and a rod") and VisIt 3.1.1 windows; "some of the slides use different data for a 3D pulsing coral. Unfortunately I don't have this data."
- The 2015 and 2020 decks begin with "Begin by reading the tutorial VISIT/MATLAB – How to make it look better" and "many of the same things apply in 3D" (in the 2021 deck the 3D part follows that tutorial): that tutorial is chapter 26 here.

Reused: the four soft-coral renderings (2015, again in 2020 and 2021) and three 2021 windows. New: every other picture, rendered with VisIt 3.5.0 (Linux, without a window) from two runs of this tutorial; each recipe in VisIt 3.5's names, with the places where they no longer work as written (slides 10, 20, 21, 27); the camera (one line in the lectures: "zoom in and modify the camera angle"), the Box operator, ThreeSlice and streamlines, which they do not cover; and `examples/analysis/visit_3d.py`, which makes every new picture. The README of IB/08 promised this chapter its ThreeSlice, contours and streamlines around the fibre.

THE TWO RUNS

	IB/08 glycocalyx (chapter 12)	IBFE/10 plate in a flow tank (chapter 22)
body	13 600 target points: one fibre, a cylinder 0.026 across and 0.156 tall on the floor	a plate $2 \times 2 \times 0.125$, 3386 nodes and 11 481 tetrahedra; a tethered band, two free flaps
box	$\frac{1}{8} \times \frac{1}{8} \times 1$, periodic in x and y; the lid slides at $V = 0.1$	$32 \times 16 \times 16$; the velocity $(0.1 \tanh 2t, 0, 0)$ on all six sides
grid	3 levels, ratio 4, finest $\Delta x = 1/512$	3 levels, ratio 4, finest $\Delta x = 1/16$
this run	`SHEAR_START = 1` (the shear flow from the first step): 20 000 steps to $t = 0.2$, 727 s on 4 cores	the chapter 22 input, run again: 5000 steps to $t = 0.5$
VisIt files	`dumps.visit`: `U`, `Omega` (vectors), `P`, `Div U`; `lag_data.visit`: `glyco_512_vertices`	`dumps.visit`: the same and the force `F`; `output.ex2`: `Mesh` with `X`, `dX`, `U`, `F` per node
frames	21 (90 MB)	21 (95 MB)

VisIt adds an expression `<vector>_magnitude` for every vector of a database (`U_magnitude`, `Omega_magnitude`, and `F_magnitude` for the plate): the lectures' `omega_magnitude`. In 3D `Omega` is a vector (chapter 26): its components are `Omega[0]`, `Omega[1]`, `Omega[2]`.

The run's log (`integral{F}` at $t = 0.2$: -0.253, continuing the README's falling curve, which stops at -0.31 at $t = 0.11$) and VisIt's metadata for both runs, printed by `visit_3d.py`.

WHAT CARRIES OVER FROM 2D

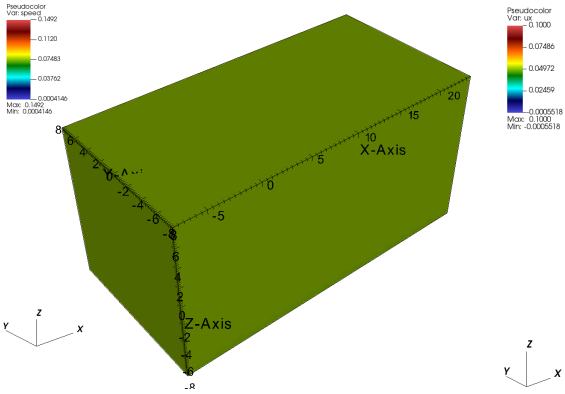
From chapter 26, unchanged in 3D:

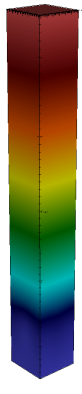
- Open `lag_data.visit` (or `output.ex2`) and `dumps.visit`; let VisIt correlate the two time sliders, or the body stays at its first frame while the fluid moves.
- **Add**, **Draw**, double-click a plot for its attributes, **Hide**/**Show**; the Subset plot of `levels` VisIt adds when it opens a SAMRAI database.
- Pseudocolor limits, nodal centring, colour tables and opacity; the Vector window's **Sampling**, **Geometry** and **Data** tabs (**Vectors**, **Glyphs** and **Data** in VisIt 3.1; the lectures' **Location**, **Form** and **Rendering**).
- **Controls** → **Annotation**, **File** → **Save window**, sessions.

New in 3D:

- A plot of a 3D field shows its outside (next slide).
- The camera has a direction, an up direction and a point it looks at, not just a zoom (slide 16).
- The operators do most of the work: Slice, ThreeSlice, Box, IntegralCurve, Displace.
- Translucency matters: an opaque isosurface hides everything inside it.

THE FIRST PICTURE: THE OUTSIDE OF A BOX





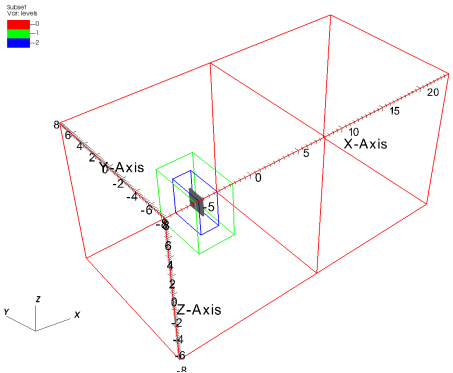
Add → **Pseudocolor**, **Draw**: VisIt draws the outer faces of the box.

- Left, the plate's tank coloured by the speed $|\mathbf{u}|$ at $t = 0.5$: the stream's speed on all six faces. The plate, 2 wide inside a box 32 long, does not show.
- Right, the glycocalyx box coloured by u_x (`U[0]`): 0.1 at the lid, 0 at the floor, the shear flow seen from outside. The fibre does not show.

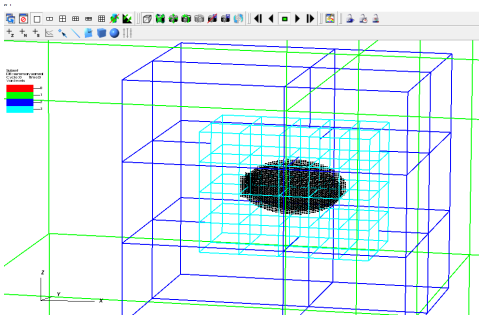
Everything inside has to be cut out (Slice, ThreeSlice, Box, Clip), turned into surfaces (Contour) or curves (IntegralCurve) first.

VisIt 3.5.0, `visit_3d.py`; axes on. The speed is the expression `speed = magnitude(U)` (slide 27).

THE REFINEMENT LEVELS IN 3D



The plate run: level 0 (red) is the whole tank, level 1 (green) a box around the plate, level 2 (blue, $\Delta x = 1/16$) a smaller box with the plate inside.

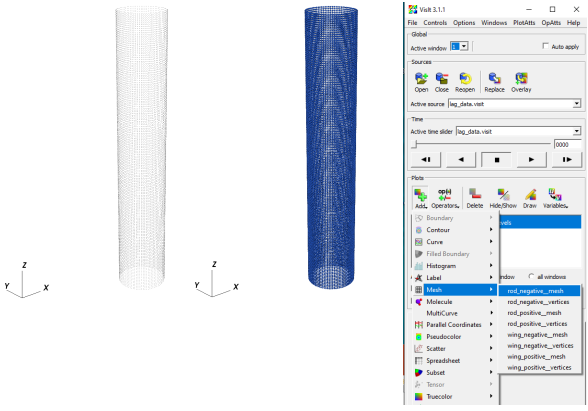


The 2021 clap-and-fling data in VisIt 3.1.1: four levels (0–3, cyan the finest), the wings' vertices in black in the middle; the legend's `summary.samrai` is the file in each `visit_dump.NNNNN` folder that `dumps.visit` lists (chapter 33), which VisIt names there; IBAMR still writes it.

The Subset plot of `levels` as a wireframe (**Subset** attributes, **Wireframe**) shows where the grid is fine, which in 3D is where a picture can show detail. **Hide**/**Show** or **Delete** it before the real plots; a script deletes it right after opening the database.

Right: the 2021 deck, slide 35 (cropped).

THE BODY AS POINTS: PIXELS OR SPHERES

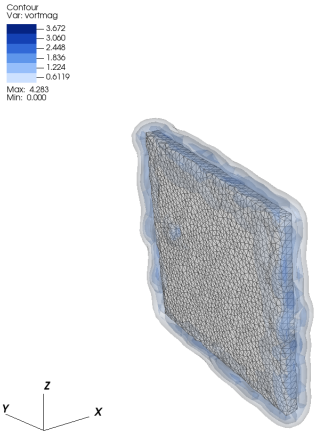


The lectures: “As in 2D: Plot vertices. Make them bigger by increasing the pixel size (use the mesh attributes to do this). You might also plot the vertices as spheres” (2015); “When loading all of the vertices, unclick the legends” (2021).

- **Add** → **Mesh** → **glyco_512_vertices**. Left: **Point type** Point, 1 pixel: the 13 600 points are a faint cylinder. Middle: **Point type** Sphere, **Point size** 0.0008 in data units, close to the points’ spacing (about 0.00097 round the cylinder and up it), so the spheres nearly touch and the cylinder looks solid.
- Right: the 2021 **Add** → **Mesh** menu of the clap-and-fling run: each of the four parts (`rod_negative`, `rod_positive`, `wing_negative`, `wing_positive`) has a `_mesh` (its springs) and a `_vertices` (its points), one Mesh plot each; in this run’s menu the suffix follows a double underscore (`rod_negative__mesh`).
- Chapter 28 colours the same points by height; its Pick slides use the points of a 2D run.

Left and middle: IB/08 at $t = 0.2$; right: the 2021 deck, slide 34 (VisIt 3.1.1).

THE BODY AS A MESH THAT MOVES: THE IBFE PLATE



- The plate is in `output.ex2`, a finite element database: **Add** → **Mesh** → **Mesh**. Its nodes are where they started; the run's displacement is the nodal variables `dX_0`, `dX_1`, `dX_2`.
- Define the vector `dX = {dX_0, dX_1, dX_2}` (**Controls** → **Expressions**, type Vector) and add **Operators** → **Transforms** → **Displace** with `dX`: the mesh is drawn where it is in each frame (chapter 28 shows the difference).
- In the Mesh attributes, **Opaque mode** On draws the faces in a solid colour, so the plate hides what is behind it; Off leaves a wireframe you can see through.
- Picture: the opaque plate at $t = 0.5$ inside six isosurfaces of $|\boldsymbol{\omega}|$ with VisIt's automatic levels (slide 14). The flaps have bent by about 0.01, too little to see at this size (chapter 22).

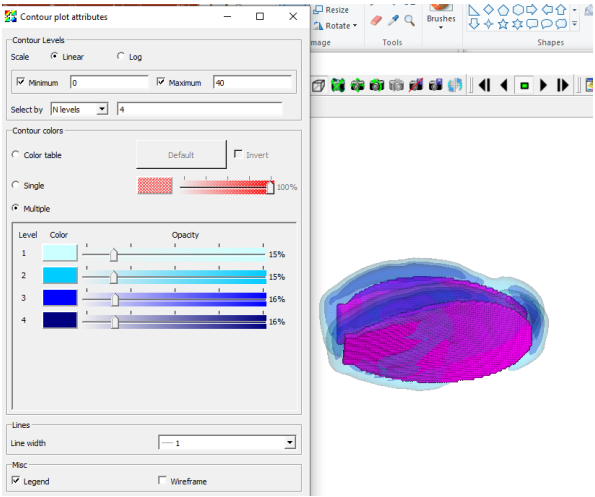
ISOSURFACES OF THE VORTICITY: THE LECTURES' RECIPE IN VISIT 3.5

the lectures (2015 wording)	VisIt 3.5
"add contour of omega_magnitude"	**Add** → **Contour** → **Omega_magnitude**, the expression VisIt makes; with a second database in the window, define `vortmag = magnitude(Omega)` yourself (slide 27)
"modify the attributes"	double-click the plot: **Contour plot attributes**
"change the minimum and maximum vorticity as in 2D to supersaturate vorticity"	tick **Minimum** and **Maximum** and type them (slide 14)
"modify N levels somewhere between 5 and 10"	**Select by**: **N levels**, and the number
"choose single for contour colors, go from lighter to darker"	**Contour colors**: **Single** (one colour) or **Multiple** (one per level, which is how lighter-to-darker is done)
"modify the color – I chose blue"	the colour button of **Single**, or each level's colour
"modify the opacity between 10% and 15%"	the opacity slider beside each colour; in a script, the fourth number of the colour (38 of 255 is 15%)
"zoom in and modify the camera angle"	slide 16

The Contour plot of VisIt 3.5 has no opacity setting of its own: the opacity belongs to the colours. A script that sets `ContourAttributes().opacity` stops with "name 'opacity' is not defined".

The 2015 slides (the 2020 and 2021 text is the same without "to supersaturate vorticity" and "go from lighter to darker"). VisIt 3.5.0.

THE CONTOUR WINDOW

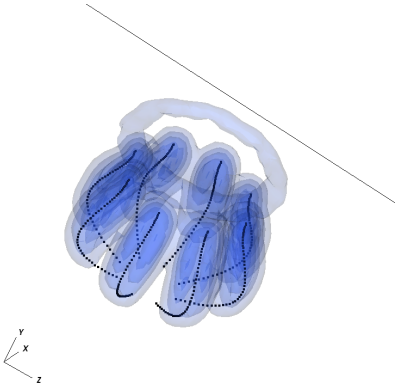


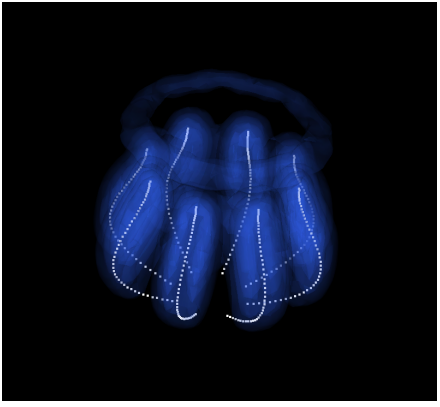
The 2021 settings for the clap-and-fling wings (VisIt 3.1.1): minimum 0, maximum 40, 4 levels, **Multiple** colours from pale cyan to navy at 15, 15, 16 and 16 % opacity. The four translucent shells wrap the magenta wing.

- **Scale**: Linear or Log spacing of the levels.
- **Select by**: **N levels**, **Value(s)** (a list you type) or **Percent(s)**.
- With **Minimum** and **Maximum** ticked, N levels run from the minimum to the maximum, both included (0.06, 0.108, . . . , 0.3 in slide 13; 0, 13.33, 26.67, 40 on the legend of the 2021 slide 37). Unticked, VisIt spreads them inside the data's range: the levels are $k\,(\max - \min)/(N+1)$ above the minimum, $k = 1, \ldots, N$ (0.612, . . . , 3.672 for a range 0 to 4.283).
- **Lines**: the width, for contours on a slice (slide 24). **Misc**: legend, wireframe.

Left: the 2021 deck, slide 38. The level values: the legends of slides 13 and 9.

WHAT THE RECIPE MAKES: THE SOFT CORAL

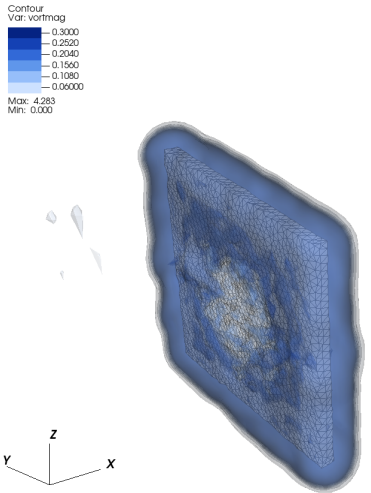




A pulsing soft coral: the tentacles as dotted lines of vertices, inside translucent blue isosurfaces of the vorticity magnitude. Left, “this simulation is on a very coarse grid”; right, the coral on black and from another angle (the view of slide 23): “Playing with the colors (controls $\rightarrow$ annotation) can modify things a lot”. The data of these pictures is not available (2021: “Unfortunately I don’t have this data”), so the recipe is redone on the plate.

The 2015 lecture, slides 6–7 (the same pictures in 2020 and 2021).

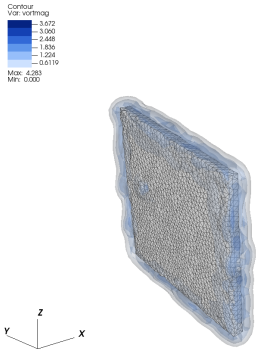
THE RECIPE ON THE PLATE



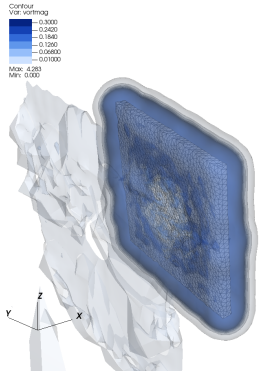
- `vortmag` $= |\boldsymbol{\omega}|$ at $t = 0.5$, 6 levels from 0.06 to 0.3, **Multiple** colours from pale to dark blue, each 15 % opaque; a Box operator first (slide 19); the plate opaque, displaced.
- The shells wrap the plate: the dark ones hug it, the pale ones stand off it. Behind the plate there is no wake of shells: the stream is $0.1 \tanh(2t) = 0.076$ at $t = 0.5$, and the Reynolds number on the plate's width, 20 for the full stream of 0.1 (chapter 22), is 15 at this moment, so the vorticity stays in a sheath around the plate.
- The few pale flakes upstream (left) belong to the lowest level, 0.06; at a minimum of 0.01 they become large ragged sheets (next slide).
- The legend's Max 4.283 is the largest $|\boldsymbol{\omega}|$ anywhere (at a grid node on the plate's back face, $x = 0.125$); 0.3 is 7 % of it.

VisIt 3.5.0, `visit_3d.py -- plate`. The camera: slide 16. The largest value and its place: a MinMax query of the chapter's first test (node 3615 of level 2, at $(0.125, 0.625, 0.1875)$).

CHOOSING THE MINIMUM AND THE MAXIMUM



automatic: 0.612 to 3.672

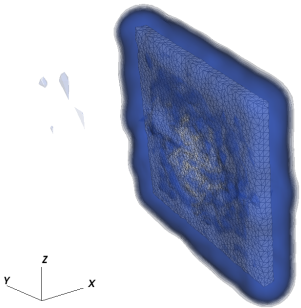


0.01 to 0.3

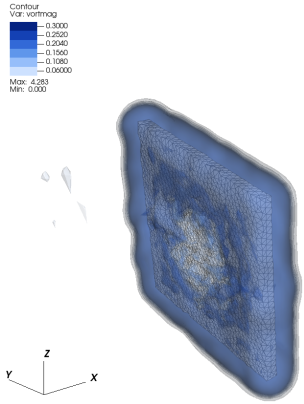
- "Change the minimum and maximum vorticity as in 2D to supersaturate vorticity" (2015). In a Pseudocolor plot a maximum below the data's saturates the colour (chapter 26). In a Contour plot the maximum is the innermost shell: everything stronger is inside it and gets no surface of its own.
- Automatic (left): the six levels are 0.612 to 3.672, all within the thin layer of strong vorticity at the plate: pale skins on the plate and nothing of the flow around it.
- Minimum too low (middle): at 0.01 the outer shell breaks into large ragged sheets upstream of the plate. Drawn with the levels' boxes (the script's 22nd picture), they lie at the upstream end of the finest level's box, where the grid changes, not in the flow round the plate: not a structure of the flow you want to show.
- Between (slide 13): 0.06 to 0.3, smooth shells that show the sheath. Try a few minima and keep the lowest that gives smooth surfaces; take the maximum well below the data's largest value.

IBFE/10 at $t = 0.5$, the same camera and colours in these two pictures and that of slide 13.

ONE COLOUR OR SEVERAL



Single: one blue, 8 levels, 12 % opaque



Multiple: 6 levels, pale to dark, 15 %

- With one colour (the recipe as the 2020 and 2021 slides wrote it) the shells add up: where they crowd, the blue is darker, so the picture reads as a soft cloud whose darkness is the number of shells in front of each other.
- With a colour per level (“go from lighter to darker”, 2015; the 2021 window) a dark shell means a high level, wherever it is.
- Either way, each shell must be transparent: at 100 % the outermost shell is all you see.

IBFE/10 at $t = 0.5$; levels 0.06–0.3 in both.

THE CAMERA

“Zoom in and modify the camera angle.”

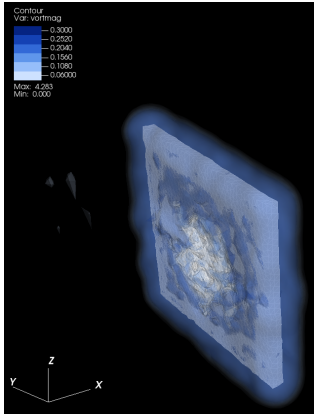
- In the window, in the **Navigate** mode of the toolbar: drag with the left button to rotate, with the middle button to zoom, with Shift and the left button to pan; the zoom mode zooms into a box you draw.
- **Controls** → **View**, tab **3D view**: **View normal** (the direction from the point looked at to the camera), **Focus** (that point), **Up Vector**, **Parallel scale** (half the height of what fits in the picture), **Image zoom**, **Perspective**.
- In these renders the automatic view kept framing the whole tank, 32 long, even after a Box (by default VisIt bases the view on the plots’ original extents: **View based on** on the **Advanced** tab, **Original spatial extents** or **Actual spatial extents**; not tried here), so the plate’s pictures set the focus just behind the plate and a parallel scale of 2.4–3.0 themselves.
- The view is saved with a session (chapter 26), so a hand-made camera can be reused. In VisIt’s Python, `print(GetView3D())` lists every setting of the current camera.

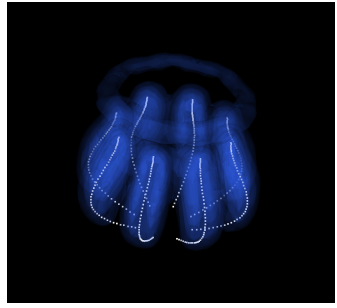
```
v = GetView3D()
v.viewNormal = (-0.55, -0.65, 0.5)  # from upstream, the
  side, above
v.viewUp = (0, 0, 1)                # z up
v.focus = (1.5, 0, 0)               # just behind the plate
v.parallelScale = 2.6               # half the picture's
  height
v.perspective = 1
SetView3D(v)
```

The camera of every plate picture with isosurfaces (slides 9 and 13–18). `viewNormal` need not be a unit vector.

VisIt 3.5.0: `View3DAttributes`.

COLOURS: THE BACKGROUND



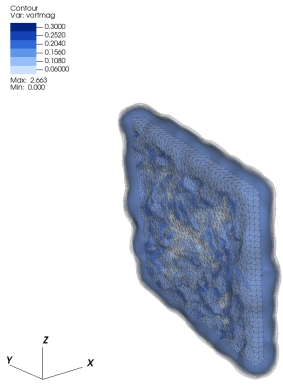


"Playing with the colors (controls → annotation) can modify things a lot."

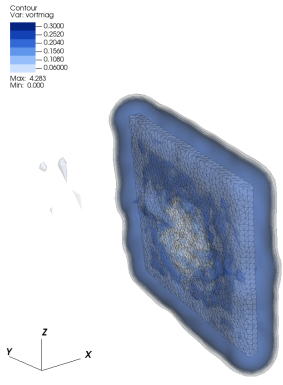
- **Controls → Annotation → Colors**: **Background color** black, **Foreground color** white (the legend's text, the axes and the triad follow the foreground).
- On black, translucent blue reads as a glow; the plate's mesh lines are drawn light grey instead of dark grey (its faces are light grey in every picture).
- For print keep white (chapter 26); for a talk black often looks better.

Left: IBFE/10, the picture of slide 13 on black. Middle: the 2015 lecture, slide 7.

THE SAME CAMERA AT TWO TIMES



$t = 0.25$



$t = 0.5$

- The stream is $0.1 \tanh(2t)$: 0.046 at $t = 0.25$, 0.076 at $t = 0.5$. With the levels fixed (0.06 to 0.3) the shells thicken as the stream speeds up; the largest $|\boldsymbol{\omega}|$ grows from 2.663 to 4.283 (the legends).
- Fixed levels are what make frames comparable, and what a movie needs (chapter 30): automatic levels would move with every frame.
- The time slider moves both sources together if they are correlated; a script sets each source's slider (`SetActiveTimeSlider`, `SetTimeSliderState`).

IBFE/10, frames 10 and 20 (counting from 0) of the 21.

THE BOX OPERATOR: ONLY THE PART YOU LOOK AT

- **Operators** → **Selection** → **Box**: ranges in x, y and z; **Some** keeps the cells that are partly inside, **All** only those wholly inside (**Amount of cell in the range**; chapter 28 has the other selection operators).
- Every fluid plot of the plate's isosurface and slice pictures had a Box first (the outside, levels, ThreeSlice and streamline pictures have none): $-2 \leq x \leq 6$, $-2.5 \leq y, z \leq 2.5$, a quarter of the tank's length around the plate. The Contour plot then has nothing to draw far from the plate.
- Operators apply in the order they are listed: Box, then Slice (or ThreeSlice). In a script, `AddOperator` adds to the plots that are selected, so add the Box right after its plot.
- The whole set of 14 plate pictures rendered in 11 s, the 8 glycocalyx pictures in 5 s.

> Cut first, then contour or slice, then set the camera on what is left.

`visit_3d.py`: the times are those of its run logs, start-up included (VisIt 3.5.0, a desktop PC under WSL).

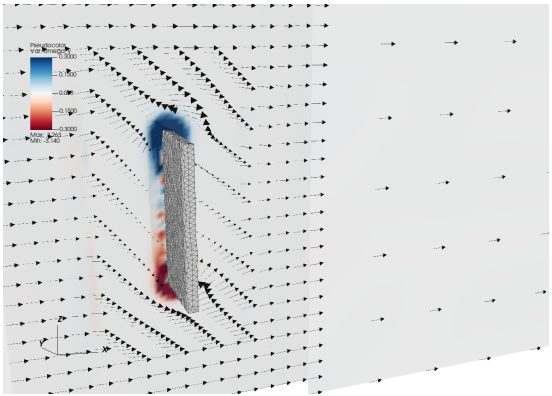
MAKING A SLICE

the lectures	VisIt 3.5
"deselect apply operators to all plots and apply subset selections to all plots in the main window"	untick the main window's **Apply operators to all plots** (and **Apply subset selections to all plots**), so that the slice goes on one plot and the body stays whole
"highlight the added Vector – U"	select that plot in the list
"choose operator attributes → selection → slicing → slice"	**Operators** → **Slicing** → **Slice** adds it; double-click it under the plot (or **OpAtts** → **Slicing** → **Slice**) for its attributes
"slice operator attributes → choose the slice you want"	**Normal**: the x, y or z axis, or any direction; **Origin**: an intercept, a point or a percentage; **Project to 2D**

- **Project to 2D** off: the slice stays in its place in 3D, with the body and the isosurfaces around it. On: a flat 2D plot, which every plot in the window must then be (slide 22).
- The plate pictures slice at $y = 0$, the plane through the middle of the plate across its two flaps; the glycocalyx at $z = -0.42$, half way up the fibre, and at $y = 0$.

The 2015, 2020 and 2021 slides (the same text). VisIt 3.5.0: `SliceAttributes` (`axisType`, `originType`, `originIntercept`, `project2d`).

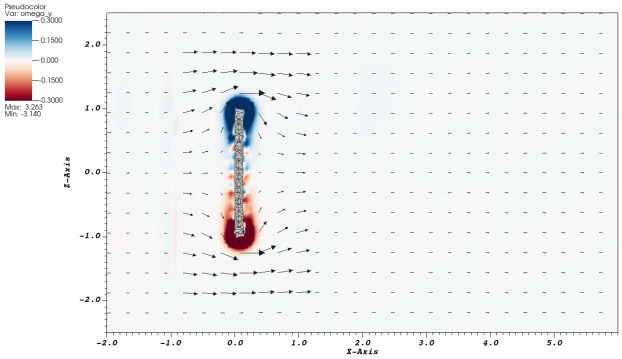
VECTORS ON A SLICE



"Add vector of U; modify the attributes; location: change the number of vectors (fixed number = 4000) and place them uniformly; form: change form quality to high; rendering: change color and set min and max values based on data."

- In 3.5: **Sampling** tab, **Uniformly located throughout mesh** and a **Fixed number**; **Geometry** tab, **Geometry Quality High** (3D arrows); **Data** tab, a **Constant** colour or a colour table, with **Limits** (VisIt 3.1 called the first two tabs **Vectors** and **Glyphs**).
- Picture: u on the slice $y = 0$ through the plate (black arrows, **Fixed number** 1500) over ω_y = `Omega[1]` at 70 % opacity, $t = 0.5$. The arrows turn round the plate's top and bottom edges.
- VisIt warned: "The option to place vector glyphs uniformly in space only works when the spatial dimension matches the topological dimension. The glyph locations will instead be a function of mesh resolution." A slice kept in 3D is a surface in space, so the arrows crowd on the finest level (left) and thin out on the coarse ones (right).

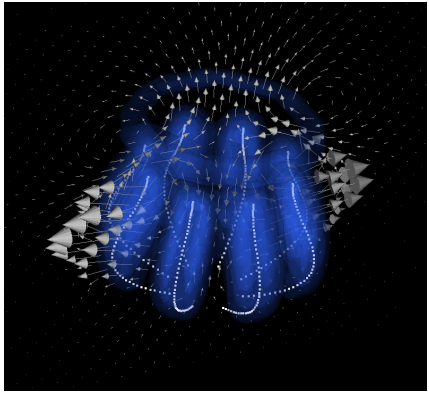
A FLAT SLICE: UNIFORM ARROWS, BUT TWO LENGTHS FOR ONE SPEED



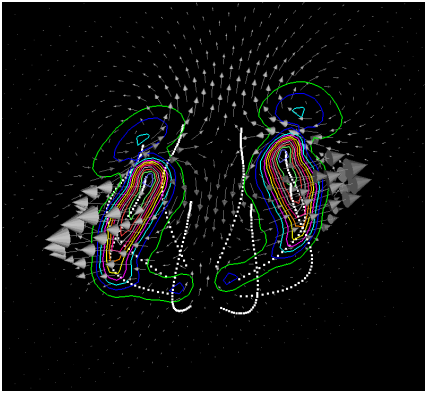
- The same three plots, every one sliced at $y = 0$ with **Project to 2D** on (the plate's mesh too, which leaves its cross-section): a 2D picture in the plane (x, z), where uniform placement works (**Fixed number** 1200; about 400 of them fall inside the Box).
- ω_y from -0.3 (red) to 0.3 (blue): positive at the top edge, negative at the bottom one, the two shear layers leaving the plate.
- Auto scale: with it on, arrows of the same speed came out at different lengths on different levels. Off, with a scale of 3, an arrow near the plate is $3|\mathbf{u}|$ long (0.23 in the stream), but on this projected slice the arrows away from the plate are still 4 times shorter, about 0.055 (compare the rows just above and below $z = 2$). On the slice kept in 3D (slide 24) the fixed scale gives one length per speed.

IBFE/10, $t = 0.5$. VisIt 3.5.0 (`VectorAttributes`: `glyphLocation = UniformInSpace`, `autoScale = 0`, `scale = 3`).

THE LECTURES' SLICES: THE SOFT CORAL



"Adding velocity vectors on a slice": white 3D arrows on a vertical slice through the middle of the coral (on the right its contour lobes run along two tentacles), with the isosurfaces.

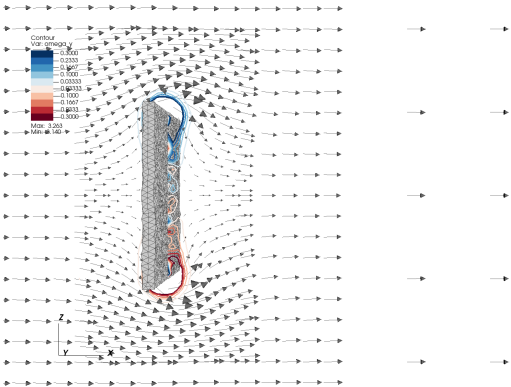


"Example of vorticity contours on a slice": coloured contour lines of the vorticity on the same slice as the arrows; the isosurfaces off.

"In similar ways you can add other properties on slices . . . Once again you just apply the operator attribute to the desired field (here I also modified some of the plot attributes of the vorticity field.)"

The 2015 lecture, slides 10–12 (also 2020 and 2021).

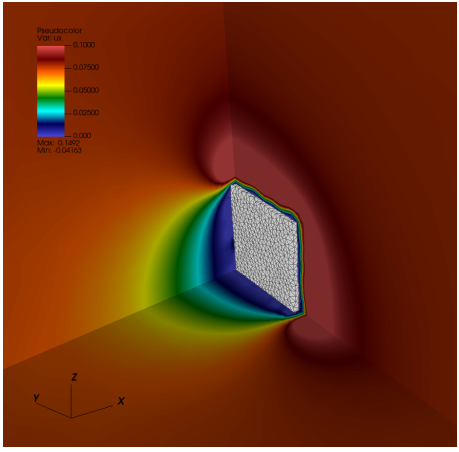
VORTICITY CONTOURS ON THE SAME SLICE



- A Contour plot of `omega_y` = `Omega[1]`: 10 levels from -0.3 to 0.3, **Color table** RdBu, lines 2 pixels wide; Box, then the same Slice ($y = 0$) as the grey arrows (**Fixed number** 2500); the camera looks along y.
- The lines close in loops round the plate's top and bottom edges, blue (positive) at the top and red (negative) at the bottom: the shear layers at the two edges.
- A Contour plot on a slice draws lines; the same plot without the slice draws the surfaces of slide 13. The operator, not the plot, decides.

IBFE/10, $t = 0.5$; VisIt 3.5.0.

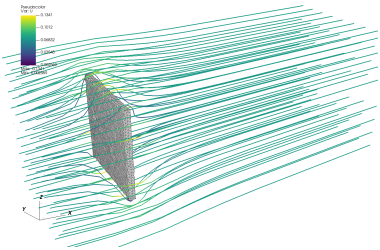
THREE SLICES AT ONCE: THREESLICE



- **Operators** → **Slicing** → **ThreeSlice**: three planes, normal to x, y and z, through one point (its attributes are the point's x, y, z). Not in the lectures.
- Picture: a Pseudocolor of u_x = `U[0]` from 0 to 0.1 on the three planes through $(0.0625, 0, 0)$, the middle of the plate, and the plate. The fluid in front of the plate is slowed (blue to green; the plane through the plate hides the slow wake behind it, chapter 22); the stream passes its edges (red).
- A quick way into a 3D field: one look shows the three directions; then one slice of the three can become a figure.
- No Box here: the planes run across the whole tank and the camera crops them.

IBFE/10, $t = 0.5$; VisIt 3.5.0 (`ThreeSliceAttributes`).

STREAMLINES: THE INTEGRALCURVE OPERATOR



- VisIt 3 has no Streamline plot (3.5's plots: Boundary, Contour, Curve, FilledBoundary, Histogram, Label, Mesh, Molecule, MultiCurve, ParallelCoordinates, Pseudocolor, Scatter, Spreadsheet, Subset, Tensor, Truecolor, Vector, Volume). Streamlines are the IntegralCurve operator, under a Pseudocolor plot.
- In a script, a Pseudocolor plot of `U` and then the operator (`AddPlot('Pseudocolor', 'U')`, `AddOperator('IntegralCurve')`). The GUI's Pseudocolor menu lists only scalars; there it is **Add** $\rightarrow$ **Pseudocolor** $\rightarrow$ **operators** $\rightarrow$ **IntegralCurve** $\rightarrow$ **U**, which brings the operator (**Operators** $\rightarrow$ **Integral Curves**) with it; in scripts that variable failed beside the plate's plot (slide 27), and the GUI was not tried. Open the operator's attributes: the seeds (here a 3×3 square at $x = -1.5$, 9×9 seeds), the direction (forward), the limits (5000 steps, steps of at most 0.05, stop after a length of 8), and what colours the curves (**Speed**).
- Picture: 81 streamlines at $t = 0.5$, coloured by the speed (0.0026 to 0.134 on the legend): they spread round the plate and close in behind it.
- These are streamlines of one frame, the flow frozen at $t = 0.5$; pathlines, which follow the flow in time, are an option of the same operator.

A TRAP OF VERSION 3.5: TWO DATABASES IN ONE WINDOW

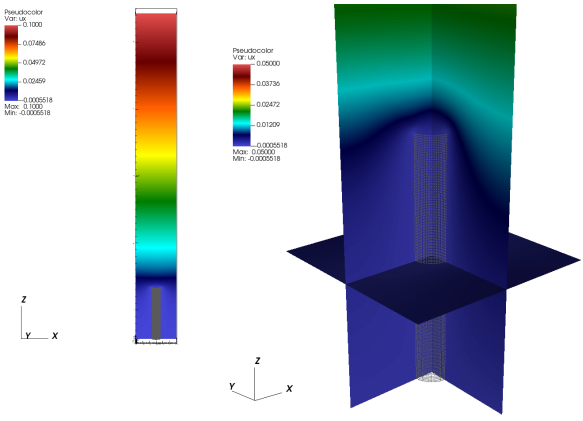
Measured with VisIt 3.5.0 (no window, the plate run). Each plot drawn alone, then with the plate's Mesh plot from `output.ex2` (and its Displace operator) in the same window:

plot of the fluid	alone	with the plate's plot
Contour of `Omega_magnitude` (VisIt's expression)	drawn	"An invalid variable (Omega_magnitude) was specified."
Pseudocolor of `U_magnitude` (VisIt's expression)	drawn (tested as a Contour)	the same error
Pseudocolor of `operators/IntegralCurve/U`	drawn	the same error
Contour of `vortmag = magnitude(Omega)` (your expression)	drawn	drawn
Pseudocolor of `U` with the IntegralCurve operator added	drawn	drawn

- So: define the expressions you need in **Controls** → **Expressions** (`vortmag`, `speed = magnitude(U)`, `ux = U[0]`) instead of using the ones VisIt makes, and, in a script, add IntegralCurve to a Pseudocolor plot of `U` rather than plotting `operators/IntegralCurve/U` (in the GUI only the latter is offered, slide 26).
- With only fluid plots in the window VisIt's expressions work, even with the plate's database open (the first two test scripts). The GUI was not tried.

Three small test scripts (`wsl/final/visit_ch27_debug*.sh` in the working folder): the time state and merely opening the second database ruled out; a plot of it in the window is what matters.

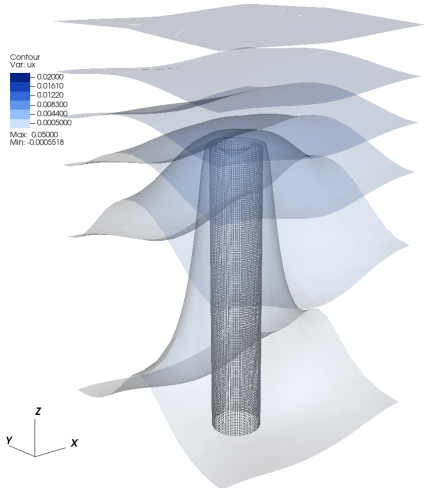
THE GLYCOCALYX: INSIDE THE BOX



- The README's steps: u_x = `U[0]`, a Slice at $y = 0$ (the x-z plane through the fibre), and the fibre's points (left). The lid drags the fluid at 0.1; the speed falls to 0 at the floor, where the fibre stands, 0.156 tall in a box 1 high.
- ThreeSlice through $(0, 0, -0.42)$, half way up the fibre, after a Box round the bottom quarter of the box (right): the horizontal plane shows the slow fluid round the fibre, the two vertical planes the shear.
- The fibre is small in its box (0.026 by 0.156 in $0.125 \times 0.125 \times 1$), so the pictures near the fibre set the focus on it and a parallel scale of 0.07–0.13.

IB/08 with `SHEAR_START` = 1, $t = 0.2$; VisIt 3.5.0.

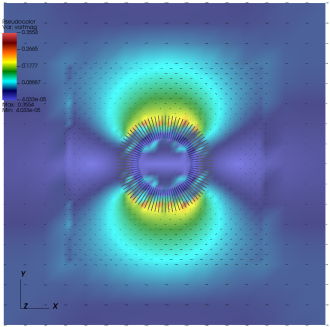
THE GLYCOCALYX: ISOSURFACES OF THE SPEED ALONG THE FLOW

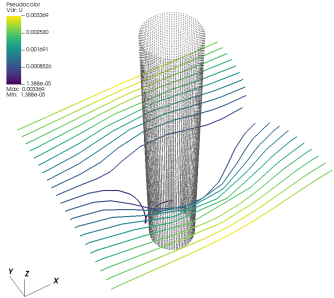


- Contour of u_x: 6 levels from 0.0005 to 0.02 (0.0005, 0.0044, . . . , 0.02), **Multiple** blues at 24 % opacity, in the bottom quarter of the box.
- In plain shear flow these would be flat, horizontal planes (u_x grows linearly with height). Near the fibre they bulge upward: the fibre slows the fluid round it, so a given speed is reached only higher up.
- Why not the vorticity, as for the plate? The shear itself has $|\boldsymbol{\omega}|$ of about $V/L = 0.1$ through most of the box, so the fibre's effect is a change on top of it (exercise 4). Contour the quantity that shows the body's effect.

IB/08, $t = 0.2$; VisIt 3.5.0.

THE GLYCOCALYX: A SLICE AND STREAMLINES AT MID-HEIGHT





- Left: the slice $z = -0.42$ seen from above, $|\boldsymbol{\omega}|$ (70 % opacity) and black arrows (**Fixed number** 3000). Beside the fibre (above and below in the picture, at $\pm y$) two bright lobes: the fluid sheared as it passes the fibre's sides.
- Right: 21 streamlines seeded on a line across the flow at $x = -0.06$, the same height, coloured by the speed. They bend round the fibre and close behind it.
- The speeds along them are at most 0.0034 (the legend), below the undisturbed shear's 0.008 at this height: the fibre, one of an endless periodic array, slows the flow near the floor (the README's note on the drag).

IB/08 with `SHEAR_START = 1`, $t = 0.2$; VisIt 3.5.0.

THE SCRIPT THAT MAKES EVERY PICTURE

```
visit -nowin -cli -s visit_3d.py -- plate <
  viz_IB3d of IBFE/10> <out>
visit -nowin -cli -s visit_3d.py -- glyco <
  viz_IB3d of IB/08> <out>
```

`examples/analysis/visit_3d.py` runs inside VisIt's Python, without a window, and writes one PNG per picture (22: the 21 new pictures of this chapter and one test of the low isosurfaces against the refinement levels) and prints a log of what it did (`ch27_render.sh` saves it).

```
ActivateDatabase(FL)
iso("vortmag", 0.06, 0.3)     # Contour, 6 levels, blues at 15 %
box()                         # Box operator on that plot
body()                        # Mesh of output.ex2 + Displace(dX)
DrawPlots(); goto(LAST)       # every time slider to the last
  frame
view((-0.55, -0.65, 0.5), focus=(1.5, 0, 0), scale=2.6)
save("ch27-plate-iso")
```

- One function per recipe: `annotations` (background, axes, legend), `body`, `iso`, `slice_op`, `vectors`, `pseudo`, `box`, `view`, `save`; each picture in `picture()`, which deletes its plots even when a step fails, so that a failed picture leaves nothing behind for the next.
- It deletes the Subset plot VisIt adds on opening, defines its own expressions (slide 27), and ends with `sys.exit()`: without it, headless VisIt waits at its prompt after the script (the first test of this chapter sat there for 25 minutes until its time limit).
- Change a camera or a level in the script, run it again: a picture takes about a second. Chapter 32 is about VisIt's Python.

WHEN THE 3D PICTURE GOES WRONG

what you see	what to do
a coloured box and nothing inside	cut or reduce the field: Slice, ThreeSlice, Box, Contour (slide 6)
the body is a speck in the middle	focus on it and lower the parallel scale, or zoom (slide 16)
one solid blob	the shells are opaque: 10–15 % opacity for each colour
ragged sheets everywhere	the lowest level is too low: raise the minimum (slide 14)
pale skins on the body and nothing else	automatic levels: set a maximum well below the data's
"An invalid variable (Omega_magnitude) was specified"	a second database in the window: your own expression (slide 27)
"name 'opacity' is not defined" (a script)	3.5's Contour has no opacity: the fourth number of each colour
"Invalid plot plugin name" for a streamline plot	streamlines are the IntegralCurve operator (slide 26)
"'VectorAttributes' object has no attribute 'UniformInMesh' " or "name 'legendFlag' is not defined" (Vector, a script)	3.5's names: `UniformInSpace`, `useLegend`
arrows crowd on the fine level	a slice kept in 3D: project it to 2D for uniform arrows (slide 22)
one speed, two arrow lengths	**Auto scale** off and a fixed scale: enough on a slice kept in 3D, not on a projected one (slide 22)
legends of an earlier picture's plots pile up (a script)	a step failed and left its plots: delete all plots after every picture
the script never ends	`sys.exit()` at the end

Every row happened while this chapter's pictures were made (VisIt 3.5.0).

EXERCISES

1. In the GUI, build the plate picture of slide 13 from the table of slide 10: open both databases, define `vortmag` and `dX`, Contour, Box, Mesh with Displace, the camera. Save the session and compare your picture with the script's.
2. Move the plate's slice to $y = 0.9$, near the plate's side edge, and to $z = 0.5$, through the top flap. How do the ω_y contours change? Which component of $\boldsymbol{\omega}$ shows the side edges best?
3. Seed streamlines on a line just above the plate's top edge instead of a square upstream. Do any of them turn back behind the plate at $t = 0.5$?
4. Glycocalyx: contour $|\boldsymbol{\omega}|$ from 0.11 to 0.3 instead of u_x. What do the shells look like now, and why (slide 29)?
5. Run IB/08 with `SHEAR_START = 0`, the course's start from rest, and make the slice picture of slide 28 at $t = 0.2$: where is the moving fluid? (The README has the 1D solution to compare with.)
6. Make the isosurface picture for every frame of the plate run with fixed levels, and then with automatic ones. Which would you put in a movie (chapter 30)?
7. The operator list also has **Isosurface** (**Operators** $\rightarrow$ **Slicing** $\rightarrow$ **Isosurface**). Put it under a Pseudocolor plot of `speed` with the isosurface variable `vortmag`: what does this show that the Contour plot cannot?

SUMMARY

- A 3D field shows its outside until you reduce it: Contour to shells, Slice and ThreeSlice to planes, IntegralCurve to curves, Box to the part you look at.
- The body: vertices as pixels or spheres (Mesh plot); a finite element mesh with Displace by `{dX_0, dX_1, dX_2}`, opaque.
- Isosurfaces of $|\omega|$: 5–10 levels between a minimum and a maximum you choose, light to dark, each 10–15 % opaque; a too-low minimum makes ragged sheets, automatic limits make skins on the body.
- The camera: view normal, up, focus, parallel scale; reuse it from a session or a script.
- On a slice: vectors (uniform only when the slice is projected to 2D; fix the scale, which evens their lengths only on a slice kept in 3D), a colour map, contour lines; one Slice per plot, applied to that plot only.
- Streamlines are an operator in VisIt 3; with a second database in the window, use your own expressions.
- `examples/analysis/visit_3d.py` makes all of this chapter's new pictures in seconds.

Next: chapter 29 (data analysis: queries, lineouts, expressions), 30 (movies), 31 (VisIt on the cluster), 32 (the command line and Python).

REFERENCES AND CREDITS

- L. A. Miller, “VISIT 3D” (2015, with Lecture 11; 12 slides) and “VISIT 3D” (2020; 11 slides); “VISIT – How to visualize IBAMR Data” (2021, slides 32–45). Pictures reused: the soft coral (2015, slides 6, 7, 10, 12), the clap-and-fling windows (2021, slides 34, 35 cropped, 38).
- VisIt: H. Childs et al., “VisIt: An end-user tool for visualizing and analyzing very large data”, in *High Performance Visualization*, CRC Press (2012) 357–372; https://visit-dav.github.io/visit-website/; version 3.5.0 (Linux), run without a window (`-nowin -cli`).
- The runs: this tutorial’s `examples/IB/08-glycocalyx-3d` (`SHEAR_START = 1`) and `examples/IBFE/10-flow-tank-plate-3d`, IBAMR 0.19.0, 4 cores.
- This chapter’s pictures and measurements: `examples/analysis/visit_3d.py`; the run and test scripts `wsl/final/ch27_*.sh` and `visit_ch27_*.sh` in the working folder.